EXPERIMENT -5

```
clc;
clear;
close all;
% Parameters
Am = 1;            % Message amplitude
Ac = 1;            % Carrier amplitude
fm = 5;            % Message frequency (Hz)
fc = 50;           % Carrier frequency (Hz)
beta = 5;          % FM modulation index
kp = 5;            % PM phase sensitivity
fs = 1000;         % Sampling frequency (Hz)
t = 0:1/fs:1;      % Time vector
% Message Signal
m = Am*cos(2*pi*fm*t);
% Carrier Signal
c = Ac*cos(2*pi*fc*t);
% FM Signal
fm_sig = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t));
% PM Signal
pm_sig = Ac*cos(2*pi*fc*t + kp*m);
% Frequency Deviation
Df = beta*fm;
% Display Parameters
fprintf('Frequency Deviation = %.2f Hz\n', Df);
fprintf('FM Modulation Index = %.2f\n', beta);

% Plot All Signals
```

```
figure;
subplot(4,1,1);
plot(t,m);
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(4,1,2);
plot(t,c);
title('Carrier Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(4,1,3);
plot(t,fm_sig);
title('FM Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(4,1,4);
plot(t,pm_sig);
title('PM Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
```

Editor

experiment005.m

/MATLAB Drive/experiment005.m

```
clc;
clear;
close all;

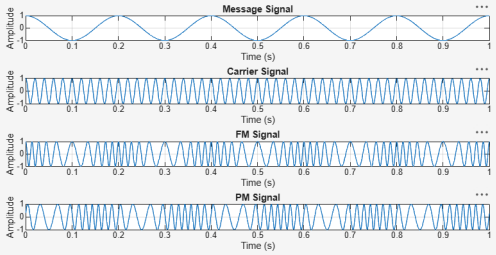
% Parameters
Am = 1;             % Message amplitude
Ac = 1;             % Carrier amplitude
fm = 5;             % Message frequency (Hz)
fc = 50;            % Carrier frequency (Hz)
beta = 5;           % FM modulation index
kp = 5;             % PM phase sensitivity
fs = 1000;          % Sampling frequency (Hz)
t = 0:1/fs:1;       % Time vector

% Message Signal
m = Am*cos(2*pi*fm*t);

% Carrier Signal
c = Ac*cos(2*pi*fc*t);
```

Figures

Figure 1



Command Window

```
Frequency Deviation = 25.00 Hz
FM Modulation Index = 5.00
>>
```